

Video Deepfake Detection for Government Communications

1. Scope

Modality: video. The brief asks for one modality done properly. This project detects AI-manipulated faces in video — face swaps and lip-sync manipulation, which is the threat model that matters for government communications (an official appearing to say something they did not).

Approach: frame-level face-crop classification with temporal aggregation. Faces are detected and cropped from sampled video frames, each crop is scored by a fine-tuned CNN, and the per-frame scores are aggregated into a single clip verdict with a calibrated confidence.

Explicitly out of scope, and stated as such in the interface:

- Audio-only manipulation. A video with authentic visuals and a cloned voice will not be caught.
 - Full-body or scene manipulation. Only facial regions are analysed.
 - Provenance verification. The tool detects manipulation artifacts, not authorship.
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2. Environment

- **Compute:** Google Colab Pro (T4 / L4 / A100 depending on assignment)
- **Framework:** PyTorch, torchvision
- **Face detection:** MTCNN via [facenet-pytorch](#)
- **Video I/O:** OpenCV, ffmpeg
- **Interface:** Gradio
- **Storage:** Google Drive for datasets and checkpoints

Total GPU time required: **under 2 hours**, including reruns. Compute is not the bottleneck; data preparation and the interface are.

3. Data

3.1 Sources

Purpose	Dataset	Notes
Training	DFDC pre-extracted face crops	224x224 crops, ~80% coverage of the DFDC training set. Requires accepting DFDC competition rules on Kaggle.
Cross-dataset test	FaceForensics++ extracted frames	Different generators and source material. Used to measure generalization, never trained on.
Demo / inference testing	DFDC <code>train_sample_videos</code> (~4 GB)	Whole videos for end-to-end pipeline testing. 77 real / 323 fake.
Demo reel	Self-generated	Public official footage plus self-generated lip-sync fakes.

The full DFDC training set (470+ GB) is not used. A subset of approximately 50,000 crops is sufficient.

3.2 Split strategy — critical

Crops are grouped by source video. Multiple crops from one video are near-identical frames.

- Split using `GroupShuffleSplit` with **video ID as the group key**.
- Every crop from a given video lands entirely in train or entirely in test.
- Where subject IDs are available, group by subject as well.
- **Never** use a random split over the image list.

A random frame split produces ROC-AUC above 0.98 that means nothing. If the first training run scores suspiciously high, verify the split before believing the metric.

3.3 Class balance

DFDC is approximately 84% fake. A model that always predicts "fake" scores 84% accuracy. Handle with class-weighted loss or undersampling of the majority class, and report precision and recall rather than accuracy alone.

4. Model and training

Architecture: ResNet-18 or EfficientNet-B0, ImageNet-pretrained, final layer replaced with a binary classification head. Fine-tune the full network — the compression and blending artifacts the task depends on are captured in early layers.

Hyperparameters:

- Optimizer: Adam, learning rate 1e-4
- Batch size: 64 (adjust to assigned GPU)
- Epochs: ~10
- Mixed precision enabled
- Input: 224x224

Augmentation:

- Horizontal flip
- JPEG compression
- Random downscale

The last two are deliberate. Real-world manipulated media arrives re-encoded and re-uploaded, not pristine. Compression augmentation is what allows the detector to survive that.

Expected runtime: 3–5 minutes per epoch on a T4; 1–2 minutes on an L4 or A100. Approximately 15–40 minutes per full run.

Colab practicalities:

- Extract the dataset to local session disk, not mounted Drive. Reading many small files from Drive can triple epoch time.
- `num_workers=2, pin_memory=True`.
- Checkpoint to Drive every epoch so a session disconnect costs one epoch, not the run.
- Run `nvidia-smi` at session start and size batches to the assigned GPU.

5. Calibration

Raw softmax output is systematically overconfident and cannot honestly be described as a confidence score.

- Hold out a validation slice from the training split.
- Fit a single temperature parameter on that slice.
- Divide logits by the temperature at inference.
- Produce a **reliability diagram** — predicted probability versus observed frequency, in ten bins — as evidence of calibration.

Output artifacts: `model.pt` and the scalar temperature value.

6. Inference pipeline

1. **Decode.** OpenCV `VideoCapture` reads the upload. Read total frame count, select ~30 evenly spaced indices, seek to each. Full decode is unnecessary.
2. **Detect and crop.** MTCNN locates the face in each sampled frame. Crop with ~30% margin around the detected box, resize to 224x224. Record how many frames yielded a detectable face.
3. **Score.** Batch all crops into a single forward pass. Apply temperature scaling. Output is an array of ~30 calibrated probabilities.
4. **Aggregate.** Count the number of frames exceeding a high threshold (~0.8) rather than taking a mean. A lip-sync manipulation leaves authentic frames between manipulated ones; averaging dilutes the signal. This heuristic outperformed averaging in the winning DFDC solutions.
5. **Verdict.** Three states:
 - **SYNTHETIC** — aggregate above decision threshold
 - **REAL** — aggregate below threshold
 - **INSUFFICIENT EVIDENCE** — fewer than roughly one third of sampled frames contained a detectable face

The per-frame probability array is retained as evidence, not discarded as an intermediate.

7. Evaluation

All metrics reported on a held-out set built with the group split described in 3.2.

Primary metrics:

- Accuracy, precision, recall
- ROC-AUC
- Confusion matrix
- Reliability diagram

Cross-dataset generalization. Train on DFDC, evaluate on FaceForensics++ frames. Report the performance drop. Most detectors report a single in-domain number and stop; the drop on unseen generators is the honest measure and is expected to be substantial.

Robustness sweep. Re-evaluate the same held-out set after simulated real-world degradation:

- Low-bitrate re-encoding
- Downscaling to 360p
- Social-media-style recompression

Report accuracy, precision, recall and AUC per condition against the clean baseline. Detectors that fail on a re-uploaded low-resolution clip are not useful against the actual threat.

Domain gap. DFDC consists of paid actors in domestic settings. The target application is government officials in press-conference and broadcast conditions. This gap is real, is expected to reduce performance, and is reported rather than concealed.

8. Interface

Built in Gradio. Designed for a non-technical user.

Primary view:

- One-sentence plain-language verdict ("This video shows strong signs of AI manipulation")
- Confidence as a word — Strong / Moderate / Weak — with the numeric probability in an expandable details panel
- Practical guidance line ("Treat as suspect. Verify against the official channel before sharing.")
- Permanent limitations notice ("Checks faces only. A real video with a cloned voice will not be caught.")

Evidence panel — "What the system noticed":

Every statement is generated from a measured value via fixed templates. No sentence is produced that is not backed by a number the pipeline actually computed.

- Region of attention, from the Grad-CAM peak: "The system focused on the mouth and jaw area, where lip-sync manipulation usually leaves traces."
- Frame count and location: "22 of 30 frames were flagged, concentrated between 0:04 and 0:07."
- Consistency: "Flagged frames were scattered rather than sustained, which lowers confidence."
- Quality caveats: "A face was detectable in only 12 of 30 frames."

Grad-CAM indicates where the model attended, not proof of manipulation. This is stated in the writeup.

9. Demo reel

8–10 samples the audience can judge before the tool answers.

- Mix of real official footage and self-generated lip-sync fakes of the same footage
 - Include at least one sample the model gets wrong, identified as such
 - All results precomputed and cached — live inference reserved for audience uploads
 - Fallback if self-generated fakes prove impractical within the timeline: build the reel from held-out DFDC clips plus real official footage, and state that the fakes are dataset-sourced
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11. Risks

Risk	Mitigation
Crop metadata does not expose source video IDs	Verify before committing to a dataset mirror. Without a group key a leak-free split is impossible. Switch mirrors.
Lip-sync generation tooling fails to install	Fall back to dataset-sourced fakes for the demo reel; state the substitution.
Colab session disconnects mid-training	Checkpoint every epoch to Drive. No run exceeds ~40 minutes.

Risk	Mitigation
Drive storage quota exceeded	Use a ~50,000 crop subset rather than the full extraction.
First run shows implausibly high AUC	Assume split leakage, not success. Verify grouping before reporting.